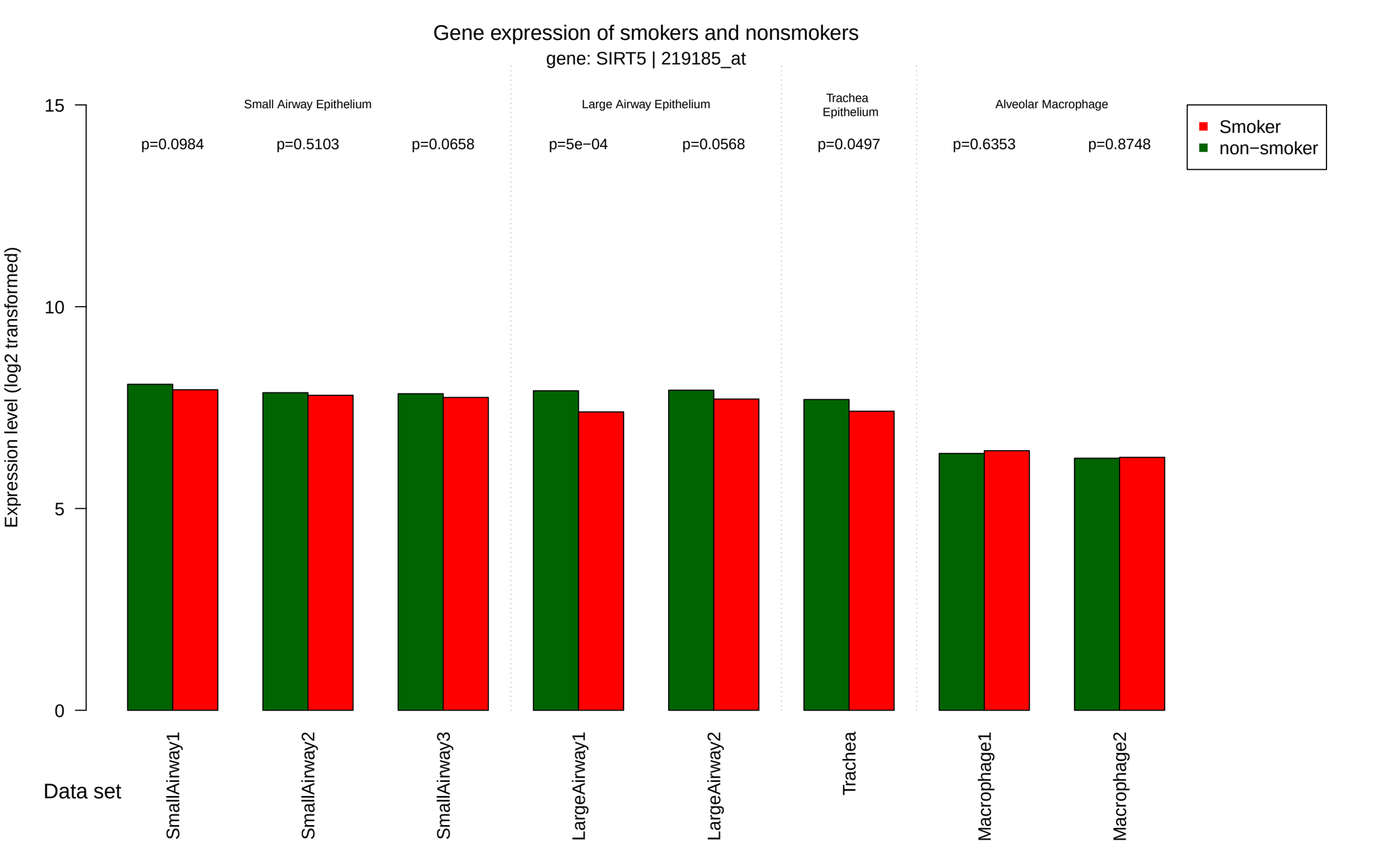
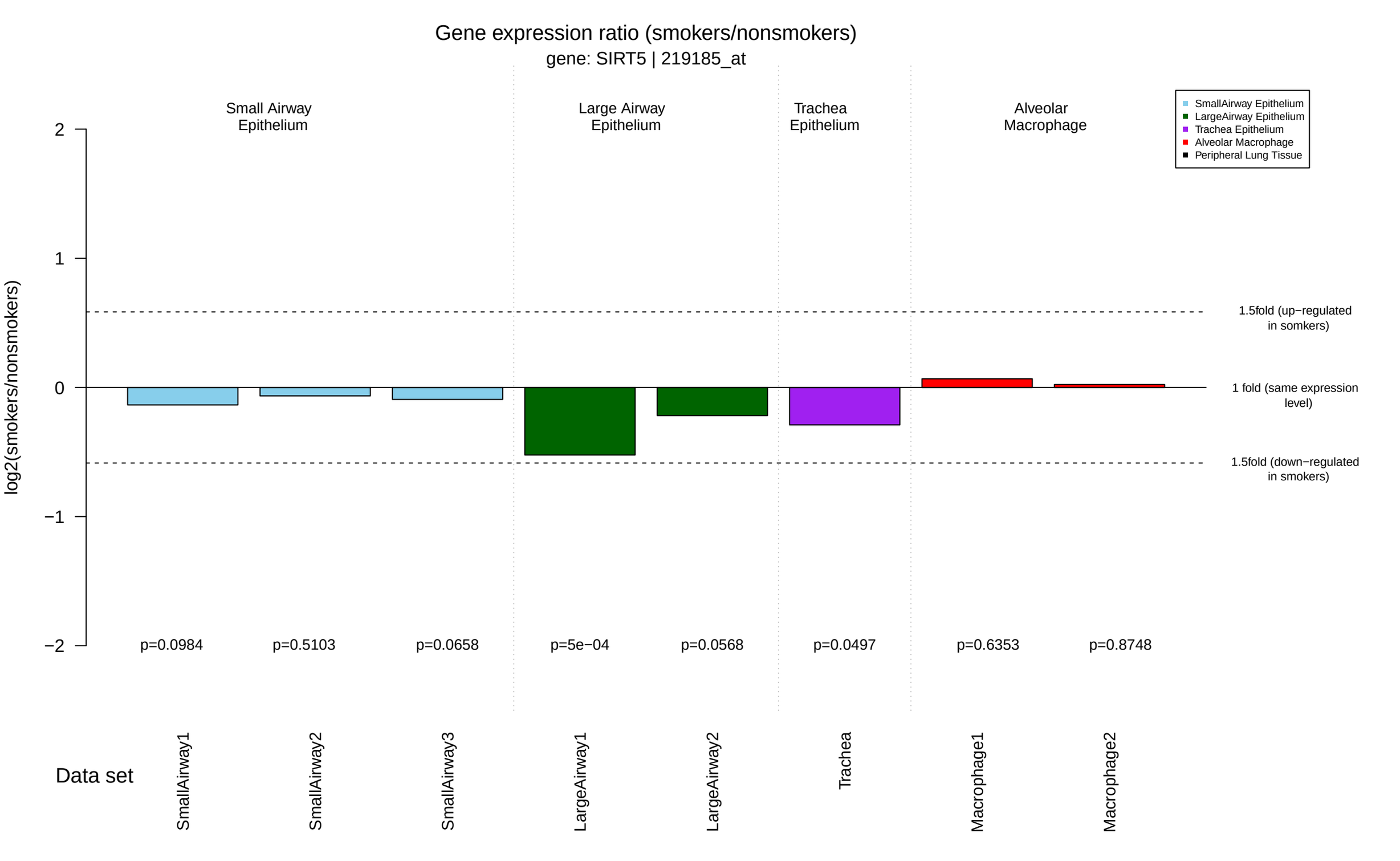
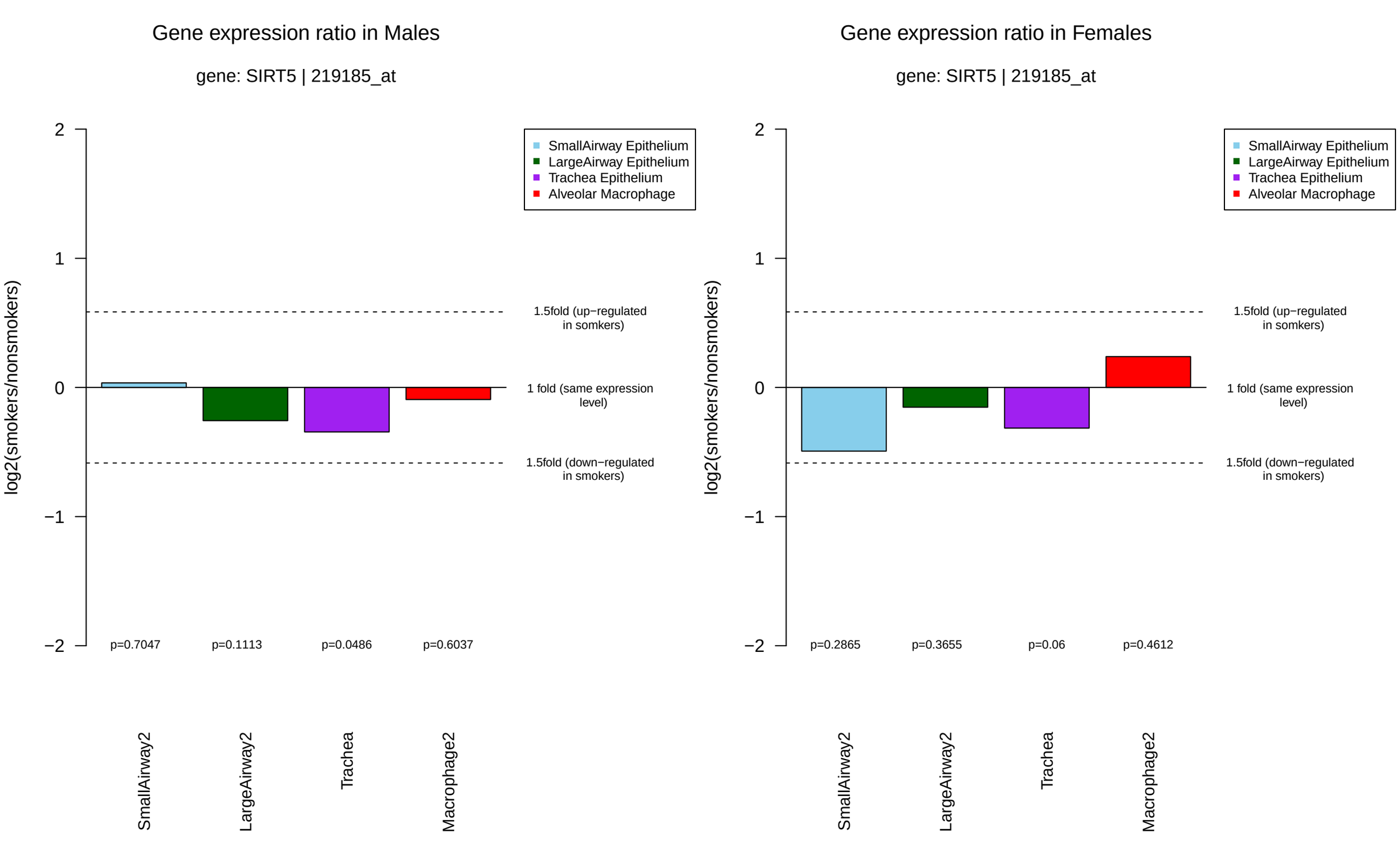
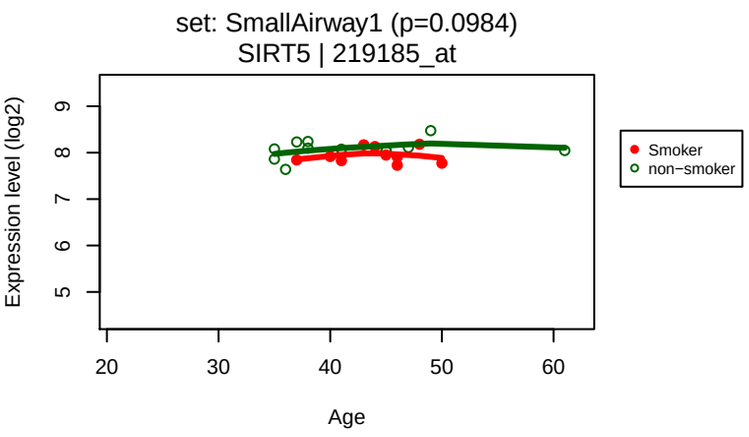
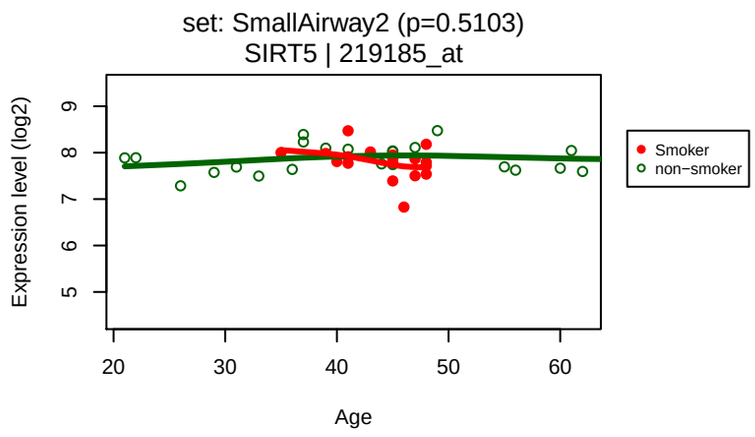
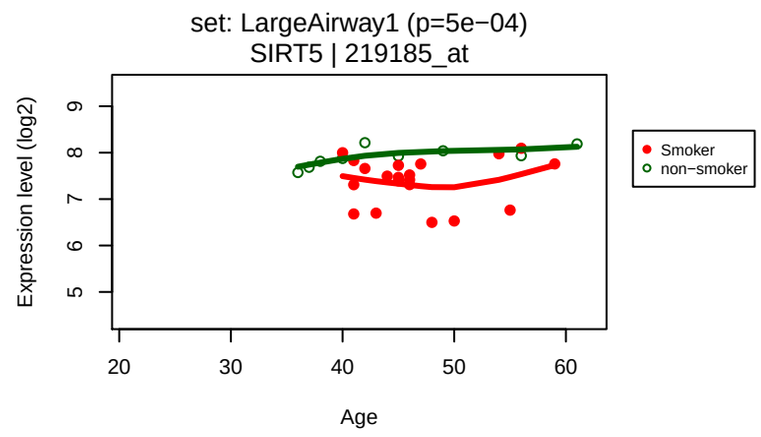

Gene expression of smokers and nonsmokers
gene: SIRT5 | 219185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0984
p=0.5103
p=0.0658
p=5e−04
p=0.0568
p=0.0497
p=0.6353
p=0.8748
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

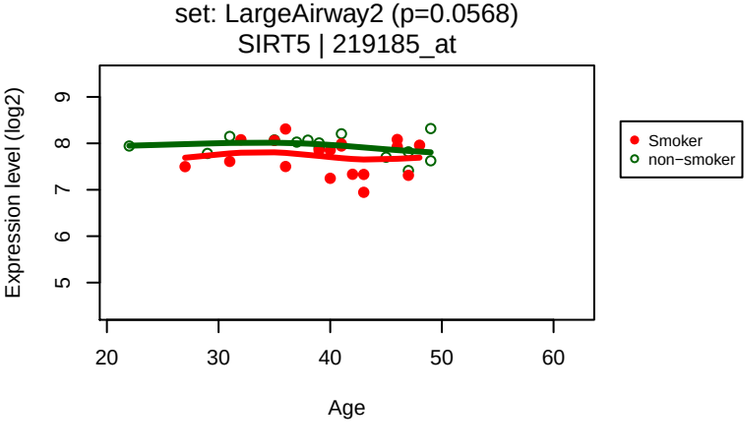

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 219185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0984
p=0.5103
p=0.0658
p=5e−04
p=0.0568
p=0.0497
p=0.6353
p=0.8748
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

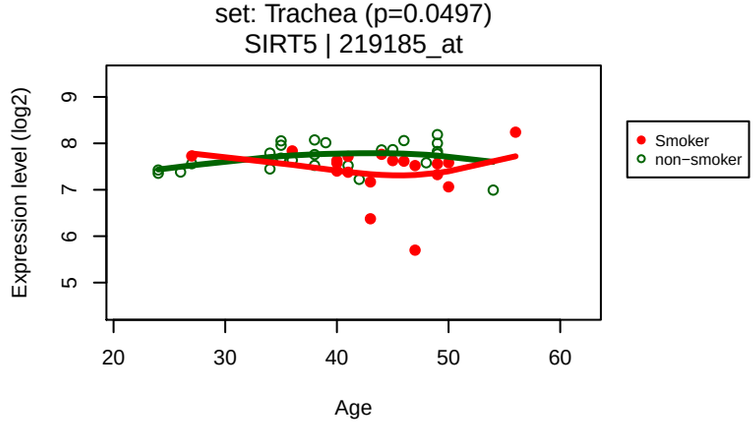

Gene expression ratio in Males
gene: SIRT5 | 219185_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7047
p=0.1113
p=0.0486
p=0.6037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 219185_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2865
p=0.3655
p=0.06
p=0.4612
SmallAirway2
LargeAirway2
Trachea
Macrophage2

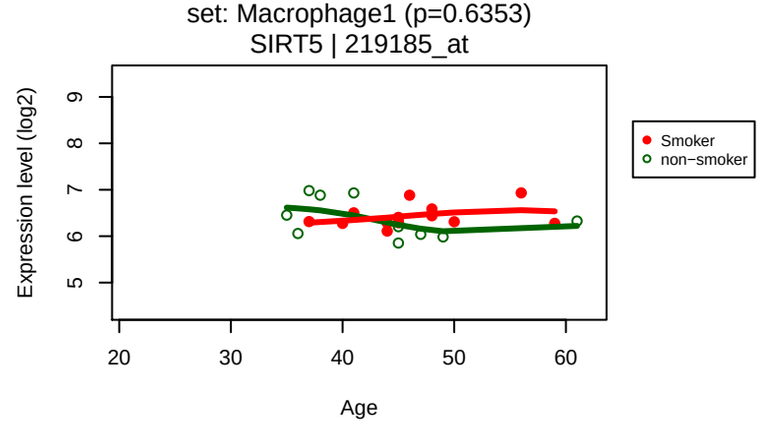

set: SmallAirway1 (p=0.0984)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker

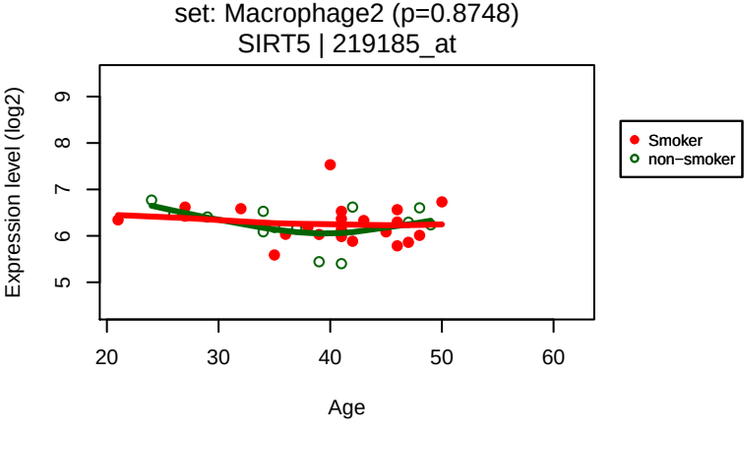

set: SmallAirway2 (p=0.5103)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker

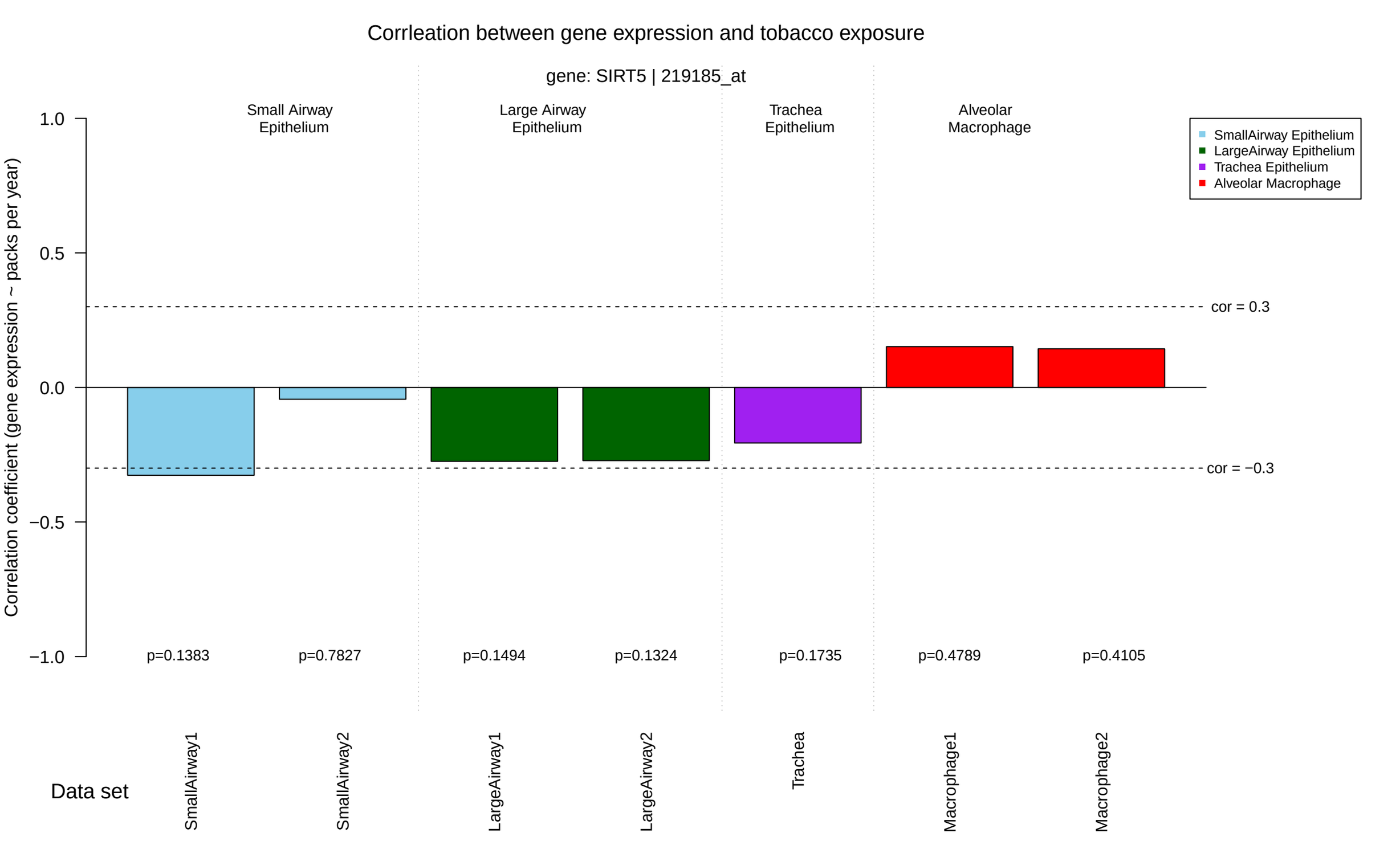

set: LargeAirway1 (p=5e−04)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0568)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0497)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6353)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8748)
SIRT5 | 219185_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 219185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1383
p=0.7827
p=0.1494
p=0.1324
p=0.1735
p=0.4789
p=0.4105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2